The ECDLP literature graph: lemmas, heuristics, proofs and where they break

Date: 2026-11-02. **Scope:** definitions, theorems, lemmas, heuristics, conjectures, counterexamples and open problems behind ECDLP attacks. Eight literature clusters plus the repository's own results, each claim linked to what it rests on.

	before	after
claims	101	**525**
typed edges	145	**888**
sources	50	**229** (83 primary text read and SHA-256 hashed)
research targets	13	**51**
refuted claims	0	**15**
claims with a Lean record	0	**29** (15 kernel-checked with a committed receipt)

`node validate.mjs` passes, `node attack-graph.mjs --check` reports 0 unlinked citations, and three claims rest on a refuted claim.

How it was built

Each cluster owned a fixed list of works, so no paper was encoded twice. Every claim records its theorem number or section, its hypotheses, and its status:

- `proved_external` only when the primary text was downloaded, read and hashed.
- `proved_external_unread` when only metadata was available (paywalls, or the eprint bot check, which was not worked around).
- `heuristic`, `conditional`, `open` or `refuted`, as the literature supports.

Each fragment validated alone; the integrator then added 25 cross-cluster edges (`cross-links.json`).

cluster	works	claims	edges	primary-read	refuted	heuristic	open
summation polynomials (`lit-sumpoly`)	Semaev, Gaudry, Diem I/II, Joux–Vitse, FPPR, FGHR, HPST, Semaev 2015, Karabina	47	86	9	1	9	3
Gröbner / fall degrees (`lit-groebner`)	BFS, Bardet, HMS, Caminata–Gorla, CCG, GMP, Salizzoni, Bigdeli et al., Steiner, Kousidis–Wiemers, Huang	48	101	13	1	2	2
prime fields (`lit-primefield`)	PKM, Amadori–Pintore–Sala,	52	90	5	1	10	6

cluster	works	claims	edges	primary-read	refuted	heuristic	open
	Semaev 2015, Kosters–Yeo, quasi-subfield polynomials, YYTK						
generic / rho (`lit-generic`)	Miller–Venkatesan, KMPT, Montenegro–Tetali, Bernstein–Lange, BLS negation, AGM, multi-DLP	42	79	10	2	6	2
Weil descent / covers (`lit-covers`)	Frey, GHS, MMT, Hess, KMPS, Enge–Gaudry, GTTD, Diem, Diem–Thomé, Smith, Joux–Vitse	57	79	11	0	10	2
lifting / transfers (`lit-transfer`)	BGJT, Cheng–Wan–Zhuang, GKZ, Kleinjung–Wesolowski, exTNFS, Barbulescu–Duquesne, Petsche, PPVW	61	91	10	5	10	5
books / surveys / standards (`lit-books`)	Galbraith–Gaudry 2016, SafeCurves, SEC 1, RFC 5639, NIST SP 800-186, Koblitz–Menezes, handbooks	47	91	5	3	11	7
this repository (`lit-repo`)	21 Lean projects and ledgers, cryptanalysis `ic-assumption-graph`	56	83	(repo)	1	4	4

The Galbraith 2018 book inventory (717 labelled results) and its promoted claims were already in the corpus.

What the graph shows

1. The degree-to-cost bridge is where binary and prime-field index-calculus estimates break

Every subexponential claim for binary-field ECDLP, and every prime-field projection, turns a degree invariant into a cost. The graph now records the evidence that this step fails:

- **First fall degree vs solving degree.** Caminata–Gorla 2021 (Ex. 4.1–4.4) put the first fall degree arbitrarily far from the solving degree, last fall degree, `d_reg` and regularity, even with field equations (`THM-GB-CG21-FIRST-FALL-INCOMPARABLE`). On summation systems specifically:
  - Kosters–Yeo prove first fall degree 2 for descended `S_3` at every `n`.
  - The degree reached grows: Kousidis–Wiemers Table 1 (4 to 5 between `n = 36` and `37`), and Kosters–Yeo at `n = 45` (degree 5).
  - PKM's own tables over prime fields climb 4, 5, 6 while the first fall degree stays 4; this repository reproduces 5 and 6.
  - Semaev 2015's own `t = 2` rows: first fall 2, solver degree 4. This is `CEX-SP-SEMAEV15-T2-TRACE`, an inference from two published results, not yet recomputed.
- **Semi-regularity.** Bardet–Faugère–Salvy conjectured that almost all `F_2` systems are semi-regular. Hodges–Molina–Schlather (Thm 7.14) prove the share tends to 0 (`CONJ-GB-BFS-SEMIREG-PROPORTION-F2`, refuted). The binary Weil-descent route and the semi-regular assumption rest on it, so the taint rule now flags both.
- **`d_reg` under-predicts the solving degree** on Gaudry-style relation systems: 5 vs 4 on 150,000 instances, and 15 vs 8 where `d_reg` is undefined.
- **Same name, different invariants.** "Last fall degree ≤ degree of regularity" holds only when `d_reg` means a solving degree; Salizzoni Ex. 3.9 gives `d_reg + 1`. Several repository files use "degree of regularity" in more than one sense (`RT-GB-DREG-READING-AUDIT`).
- **The sound routes are attached to `GAP-FALL-DEGREE-COST-BRIDGE`:**
  - Caminata–Gorla Thm 3.1: `sd = max{last fall, max GB degree}`.
  - Salizzoni: `sd ≤ d_reg + 1`.
  - CG: `sd ≤` the CM regularity in generic coordinates.
  - CCG: regularity of Weil restrictions grows exactly linearly in `n`.

Nagao's Lemma 4 (the starting point of this audit) is one instance of this pattern, not the only one.

2. The rho baseline is itself partly heuristic

- Every rigorous rho theorem assumes a uniformly random partition (`ASM-GG-RANDOM-PARTITION`), which implementations do not use.
- No rigorous square-root bound is known for the r-adding walks used in practice (`GAP-GG-ADDING-WALK-RIGOROUS`).
- The best proved constant is about 52.5·√|G|, against the heuristic 1.2533.

So "matched rho" references rest on a heuristic constant plus measurement (`GAP-GG-RHO-CONSTANT qualifies FACT-MEASURED-BOUND`). In addition, Bernstein–Lange show the Brent–Pollard correction `1/√(1−1/r)` undercounts (r = 6: 1.129 vs 1.095), and `ecc2k130/WALK-CONSTANT.md` uses the Brent–Pollard form.

3. Transfers: heuristic vs proved

- Every L(1/3) NFS, TNFS and exTNFS cost rests on a smoothness heuristic (`HEUR-XT-NFS-SMOOTHNESS`, 8 claims downstream). No target-field cost has a proved lower bound.
- In fixed characteristic, Kleinjung–Wesolowski prove quasi-polynomial DLP, so MOV on small-characteristic supersingular curves is provably quasi-polynomial.
- Several pairing-security claims are refuted by later estimates or breaks: BN-3072 at 128 bits, the genus-2 supersingular curve over F_2^367, and AMOR F_2^1223.

4. Standards disagree, and none derive their thresholds

	embedding degree	**CM discriminant / class number**
SEC 1 v2	B ≥ 100	none
NIST SP 800-186	k ≥ 2^10	none ("no technical rationale")
Brainpool (RFC 5639)	(q−1)/k < 100	class number > 10^7

	embedding degree	CM discriminant / class number
SafeCurves	≥ (ℓ−1)/100	|D| > 2^100

SafeCurves misquotes SEC 1 (it says 20) and Brainpool (it says 10^6) against the current texts. Both are recorded as refuted against SEC 1 v2 and RFC 5639; older versions may explain the difference.

Repository findings (to fix in follow-ups, not in this PR)

file	finding	verified here
`src/ecc_safety.rs:211-215`	Amortised rho is attributed to “Galbraith-Lin-Scott” (Kuhn–Struik is the usual attribution), and described as helping an attacker who wants *any one* of m keys. No speedup is known for the first of m keys; the √(n/m) is per key when breaking all m. The computed margin is valid under the correct reading.	yes (source read)
`src/cryptanalysis/ghs_descent.rs:45`, `problems/ghs-higher-magic-descent-not-executable.md:37`	Cite “Hess 2004, Theorem 5.3”. Hess numbers theorems consecutively, so there is no 5.3; the injectivity result is Theorem 14.	yes (Hess text)
`src/cryptanalysis/jacobian_transfer/cost.rs:270-278, 376-425`	GTTD q^(2-2/g) applied at every genus (proved for g ≥ 3 with conditions); Enge–Gaudry L(√2) hypothesis “reported, not enforced”; gGHS covers are non-hyperelliptic, so their cost is open.	lines confirmed
`research/transfer_audit_20261011/PROTOCOL.md`	Calls L[1/3, (32/9)^(1/3)] an “attacker-favourable lower bound”; it is unproved, and below the proved quasi-polynomial bound in characteristic 2 and 3.	agent report
`research/factor_base_arity_review_20261024/lit_degree_reduction.md` vs `papers/ic-negative-results` S6	One says Semaev’s Assumption 1 is	agent report

file	finding	verified here
	refuted at n = 45; the other says the off-diagonal failure is not established.	
`src/cryptanalysis/preprocessing_rho.rs`	Table described as N^{2/3}; in Bernstein–Lange the table holds ~ℓ^{1/3} distinguished points covering ~ℓ^{2/3} points. Worth checking the cost model.	agent report
five Lean “kernel-checked” claims	No receipt in the tree (four JKSST counterexamples, Koblitz `CyclicWindow`); recorded as unchecked.	yes

Where to attack next

The full ranked frontier (318 claims) is in ATTACK_GRAPH.md. The claims carrying the most weight:

claim	status	rests on it	settle by
`HEUR-DECOMPOSITION-PROBABILITY`	heuristic	13	exact counts vs 1/(ng)! on the families in use (`RT-BK-DECOMP-MEASURE`)
`HEUR-RELATION-RANK`	heuristic	10	rank deficiency measured at scale
`ALG-WD-GAUDRY-2000`	conditional	9	read the primary text (paywalled)
`HEUR-XT-NFS-SMOOTHNESS`	heuristic	8	not settleable here; keep transfer costs labelled heuristic
`COND-ECC2K130-FFDA-PROJECTION`	conditional	8	`RT-GB-SUMMATION-INVARIANTS`: measure all five invariants on S_3/S_4 descents as n grows
`HEUR-FIRST-FALL-DEGREE`	heuristic	7	as above; Caminata–Gorla route for any cost row
`HEUR-RHO-RANDOM-MAPPING`	heuristic	7	`RT-GG-WALK-NONRANDOMNESS`

Cheap, decisive targets that are ready now:

- `RT-SP-T2-TRACE-CHECK`: confirm or kill the Semaev 2015 `t = 2` counterexample by exact computation.
- `RT-BK-ONE-OUT-OF-L`: correct the multi-target model in `ecc_safety.rs`.
- `RT-WD-HESS-KERNEL-DESCENT`: fix the Hess citation and use Theorem 14 as the injectivity certificate.
- `RT-PF-APS-VS-PKM-REMEASURE`: the APS “beats PKM” comparison sets its m = 3 runs against PKM’s m = 2.
- `RT-LEMMA4-REVERSE`: the outside form of `d'_F ≤ d_F`.

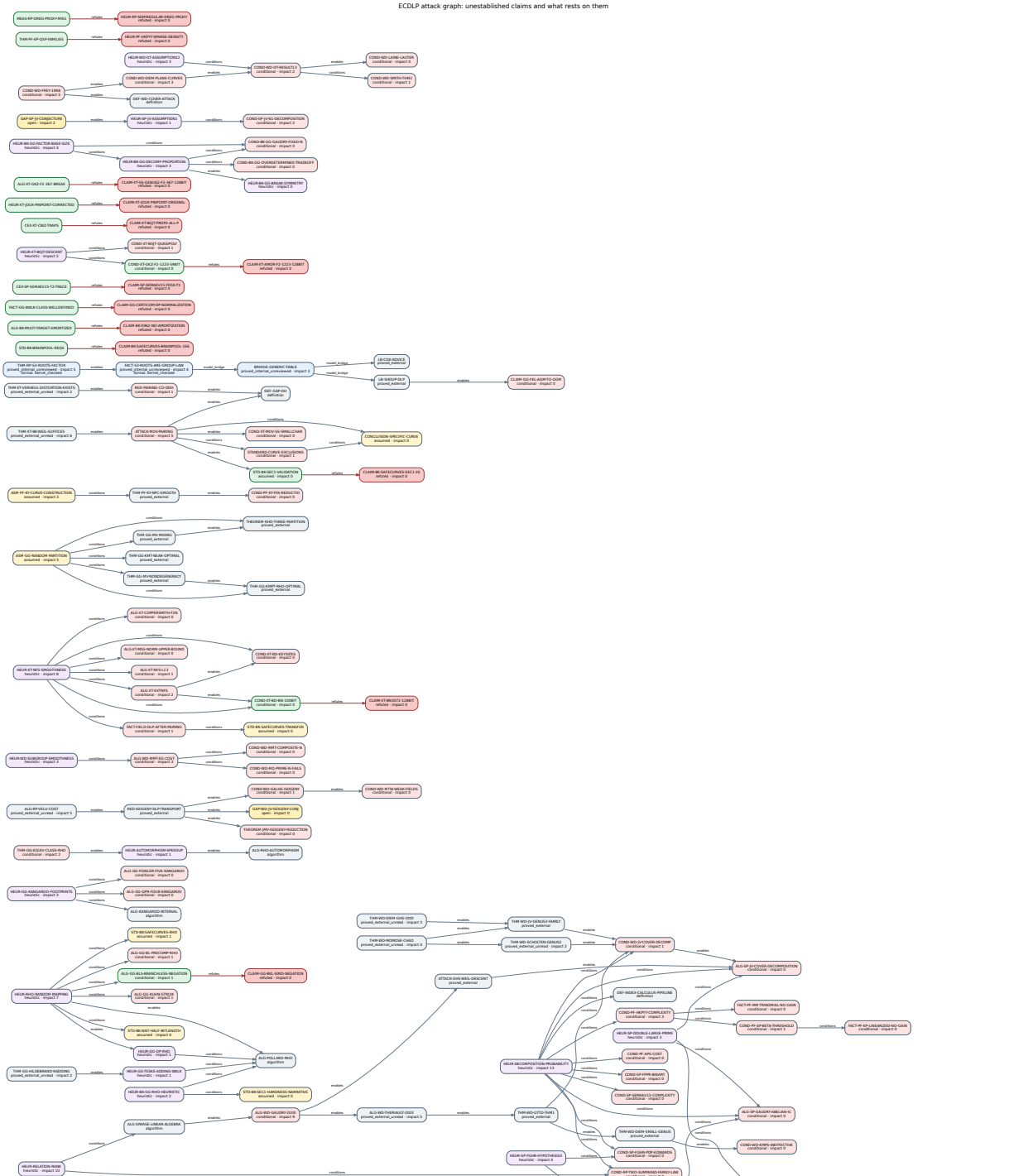
Limits

- 64 sources are metadata-only and 35 secondary-only, so their 30 theorems are `proved_external_unread`. Paywalled books (the Cohen–Frey handbook, Hankerson–Menezes–Vanstone, Washington, Blake–Seroussi–Smart) are located to the chapter only.
- eprint PDFs were not fetched (bot check). Unread as a result, among others: Diem's Math. Comp. paper and Diem 2013 beyond its sample pages, Gaudry 2000, Adleman–DeMarrais–Huang, Menezes–Qu, Petit–Quisquater, HKY 2015 and Semaev–Tenti.
- Agent-reported findings marked "agent report" above were not independently re-derived by the integrator.
- The first-fall-degree assumption stays `heuristic` rather than `refuted`: the general form is refuted (Caminata–Gorla), but the estimates use a family-specific form, for which the evidence is strong but not a proof.

Files

- Fragments: `lit-sumpoly.json`, `lit-groebner.json`, `lit-primefield.json`, `lit-generic.json`, `lit-covers.json`, `lit-transfer.json`, `lit-books.json`, `lit-repo.json`, `cross-links.json`, plus `fall-degree-lemmas.json` (the Lemma 4 audit, `LEMMA_LINKAGE.md`).
- Generated: `ATTACK_GRAPH.md`, `attack-graph.{json,dot,svg,pdf}`, `graph.{dot,svg}`, `overview.{dot,svg,pdf}`, `claims.tsv`, `sources.tsv`, `stats.tsv`.
- Report PDF: `LITERATURE_GRAPH.pdf` (this file plus the attack-graph figure).

Figure: attack graph (claims carrying at least two others, and every refuted claim)



Attack graph